

TOC and Alkalinity in Drinking Water Treatment

What is Total Organic Carbon (TOC)?

Total Organic Carbon (TOC) is a measure of the amount of organic material present in water. Sources include decaying vegetation, algae, microorganisms, natural organic matter, and other carbon-containing compounds found in source waters. In drinking water treatment, TOC is an important indicator of the effectiveness of coagulation, flocculation, sedimentation, and filtration processes used to remove natural organic matter. TOC is particularly important because organic matter reacts with disinfectants such as chlorine to form disinfection byproducts (DBPs). The primary regulated DBPs are Total Trihalomethanes (TTHMs) with an MCL of 80 µg/L and Haloacetic Acids (HAA5) with an MCL of 60 µg/L. Denver Water typically observes source water TOC concentrations between 1.5 and 4.0 mg/L.

What is Alkalinity?

Alkalinity is a measure of water's ability to neutralize acids and resist changes in pH. It is primarily derived from bicarbonate, carbonate, and hydroxide ions and serves as the buffering capacity of water. During coagulation, optimal TOC removal typically occurs under slightly acidic conditions. Higher alkalinity can make it more difficult and costly to lower pH into the optimal treatment range because additional chemicals may be required. Alkalinity is also critical for finished water stability and corrosion control within the distribution system.

Why Are TOC and Alkalinity Important to Predict?

TOC and alkalinity are among the most dynamic and operationally significant water quality parameters affecting treatment performance. A key regulatory threshold occurs at 60 mg/L alkalinity. When source water alkalinity is below 60 mg/L, treatment plants are generally required to achieve 35% TOC removal; above 60 mg/L, the required removal decreases to 25%. Seasonal fluctuations around this threshold increase treatment costs, operational complexity, and the risk of elevated DBP formation during runoff and storm events. Improved forecasting and predictive modeling would allow operators to proactively adjust treatment strategies, optimize chemical usage, and reduce regulatory risk.

Conclusion

TOC and alkalinity are critical parameters that influence treatment effectiveness, regulatory compliance, operational costs, and drinking water quality. For Denver Water, improved prediction of these parameters can enhance operational decision-making, optimize treatment performance, and support the continued delivery of safe and reliable drinking water.

Helpful links:

How treatment works:

<https://www.youtube.com/watch?v=5w93Mb4EIVM&list=PLiwQcGPBWz0lbnbe8JbEH1T7M9gKdevxB&index=5>

Treatment Process walk through: <https://www.denverwater.org/your-water/treatment-process>

Quick Guide: DBP Precursors rule, regulation 11, CDPHE

<https://drive.google.com/file/d/0BwDv77AW5PLkNTFaWUhfMEl1aXc/view?resourcekey=0-lMtFl7550yeyz1m8kloesQ>

Disinfection Byproducts Fact sheet, CDPHE

https://docs.google.com/document/d/14POqfo5xTZk0DXnfQ78hnE7B4yiKpY-M_WC7HeZOKGQ/edit?tab=t.0

Disclaimer:

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